

NAME: _____

MATRIC NUMBER: _____

SCHOOL OF MATERIALS SCIENCE & ENGINEERINGCONTINUOUS ASSESSMENT 2 SEMESTER 1 2023-2024MS1016 – THERMODYNAMICS OF MATERIAL

14 November 2023

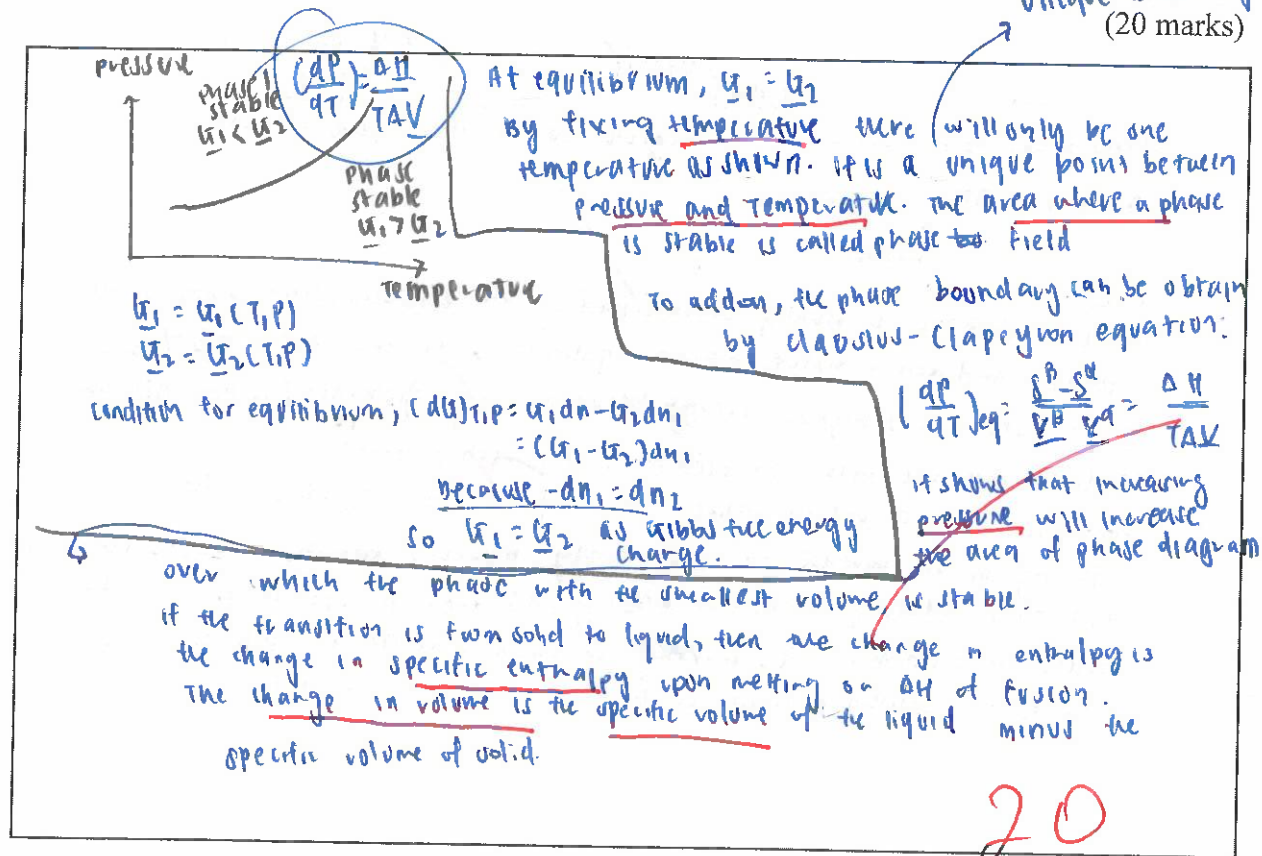
Time Allowed: 1 hour

INSTRUCTIONS

1. This paper contains **THREE (3)** short questions and **TEN (10)** MCQ questions.
2. Answer **ALL** questions.
3. This is an open book test.
4. All questions have **ONLY ONE** correct answer for the MCQ questions.

PART I – SHORT QUESTIONS

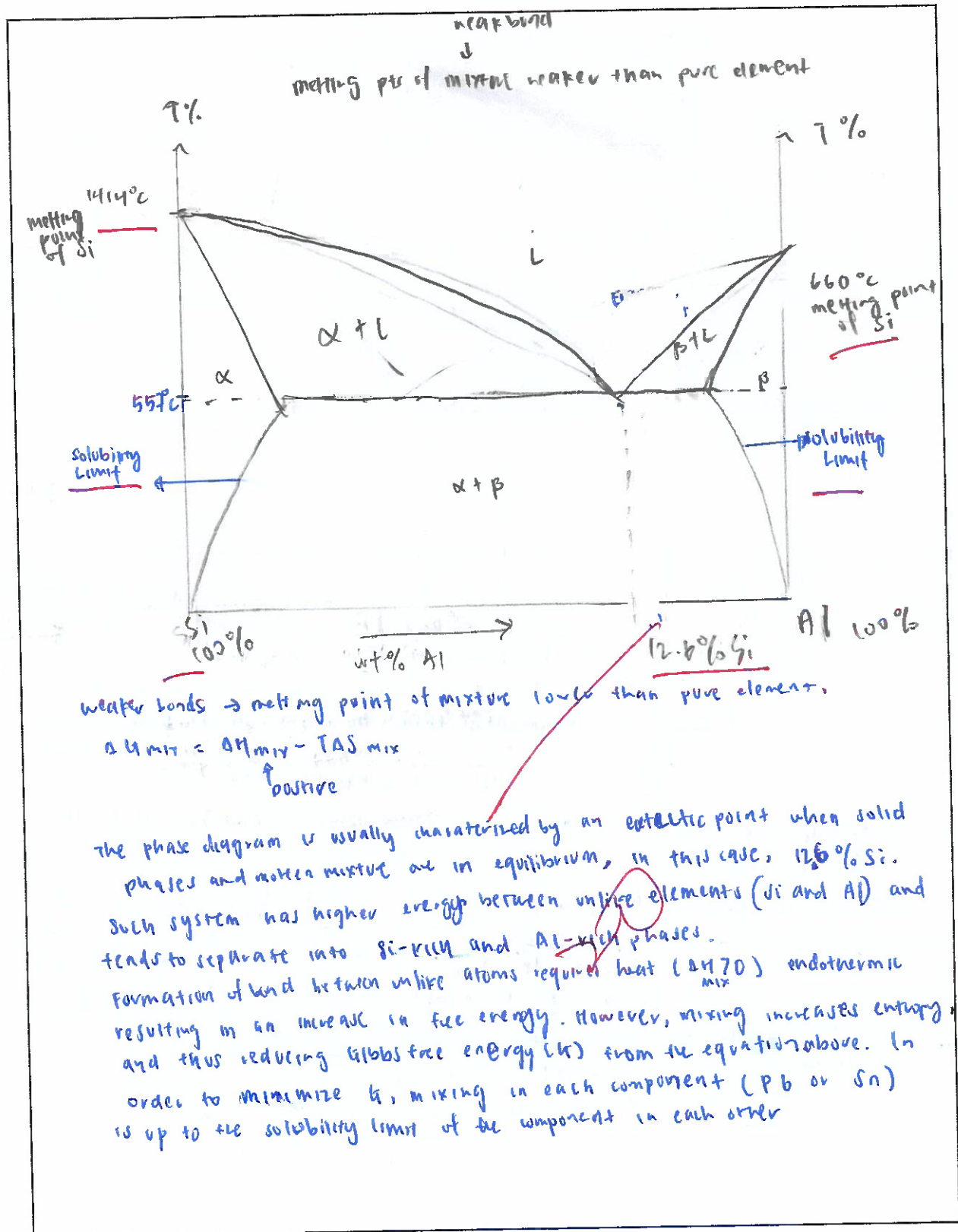
- (1) Explain the important factors that affect the equilibrium conditions between phases of a single component system. (20 marks)



don't mix well

2. The enthalpy of mixing, ΔH_{mix} of Al and Si is positive. Given that the melting point of Al and Si are 660°C and 1414°C respectively, sketch the equilibrium phase diagram for Al and Si binary system. At 557°C , the solid phases and the molten mixture are in equilibrium, having a composition of Al 12.6%Si,

(20 marks)



3. Ti is extracted from TiO_2 . Use the Ellingham diagram to suggest the optimum conditions such as temperature, equilibrium O_2 pressure and the appropriate reducing agent suitable for the reduction process.

(20 marks)

Carbon in the form of ~~CO~~ CO is most suitable and as a gas, it can react readily with TiO_2 during the reduction process. Inspecting the Ellingham diagram, the minimum temperature for the reduction of TiO_2 is 1650°C .

From the Ellingham diagram, the equilibrium pressure for the oxidation of TiO_2



is approximately 10^{-16} atmosphere.

Additionally, at any temperature, the oxide with lower standard Gibbs free energy of the oxidation is more stable. Therefore by inspecting the Ellingham diagram, any of the oxides with the standard Gibbs free energy lower than TiO_2 can be used to extract Ti of TiO_2 . But the best one is still carbon in the form of CO due to the impractical use of metal.

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PART II MCQ QUESTIONS

- (4) Answer all the **TEN MCQ** questions below. There is only **ONE** answer for each question.

(40 marks)

- (i) Which of the following elements will NOT oxidize at 1000°C at an oxygen pressure of 10^{-5} atmosphere? *not to oxidized*

- A Al
- B Cu
- C Mg
- D Si
- E Ti

ANS: **B**

- (ii) The melting point of a certain solid was found to decrease when an external pressure is applied. Which of the following is true?

$$\frac{dp}{dT} = \frac{H}{T\Delta V}$$

melting point *Solid → liquid*

- A the molar volume of the solid material contract during melting
- B the material will melt at higher temperature
- ☒ C the molar volume of the solid material will expand upon melting
- ☐ D the molar volume will remain constant during phase transformation
- ☐ E the melting point is constant

ANS: **A**

- (iii) [↑]High internal vapour pressure in crystals in a single-component solid with homogeneous grain will result in:

- A recrystallisation
- B nucleation within the grains
- C equilibrium
- D grain growth
- E reduction in grain size

ANS: **D**

(iv) Which of the following information of an oxidising reaction CANNOT be obtained from the Ellingham diagram?

- A equilibrium O_2 pressure for a given temperature
- B entropy of reaction
- C standard free energy of oxidation
- D enthalpy of reaction
- ☒ E rate of oxidation/reduction

ANS: E

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(v) The definition of entropy of mixing is defined as:

- ☒ A difference in the total entropy before and after the components come into contact
- B total entropy of all the components before the components come into contact
- C total entropy of all the components after the components come into contact
- D total accessible microstates of a mixture
- E total entropy of the mixture

ANS: A

(vi) Order-disorder grain microstructure is most likely to be associated with:

- A limited solubility of the components in each other
- B homogeneous grain structure
- ☒ C enthalpy of mixing is negative
- D enthalpy of mixing is positive
- E entropy of mixing is negative

ANS: C

$\Delta H \downarrow$

(vii) What is NOT an intermetallic compound?

- A FeCo
- B AlN
- C NiAl
- D CuAu
- ☒ E Fe₃C

ANS: E

(viii) Gibbs free energy is unique at constant.....

$$dU = VdP + TdS$$

- A T and V
- ☒ B P and T
- C H and P
- D T and S
- E P and V

ANS: B

(ix) When three phases of a single component are in equilibrium, which of the following statement is true

- ☒ A The molar Gibbs free energies of the phases are equal, and any amount of the phases can exist together.
- B The Helmholtz free energies of the phases are equal, and any amount of the phases can exist together.
- C The Gibbs free energies of the phases are equal, and only equal amounts of the phases can exist together.
- ☒ D The partial molar Gibbs free energies of the phases are equal, and any amount of the phases can exist together.
- E The chemical potentials of the 3 phases must be equal.

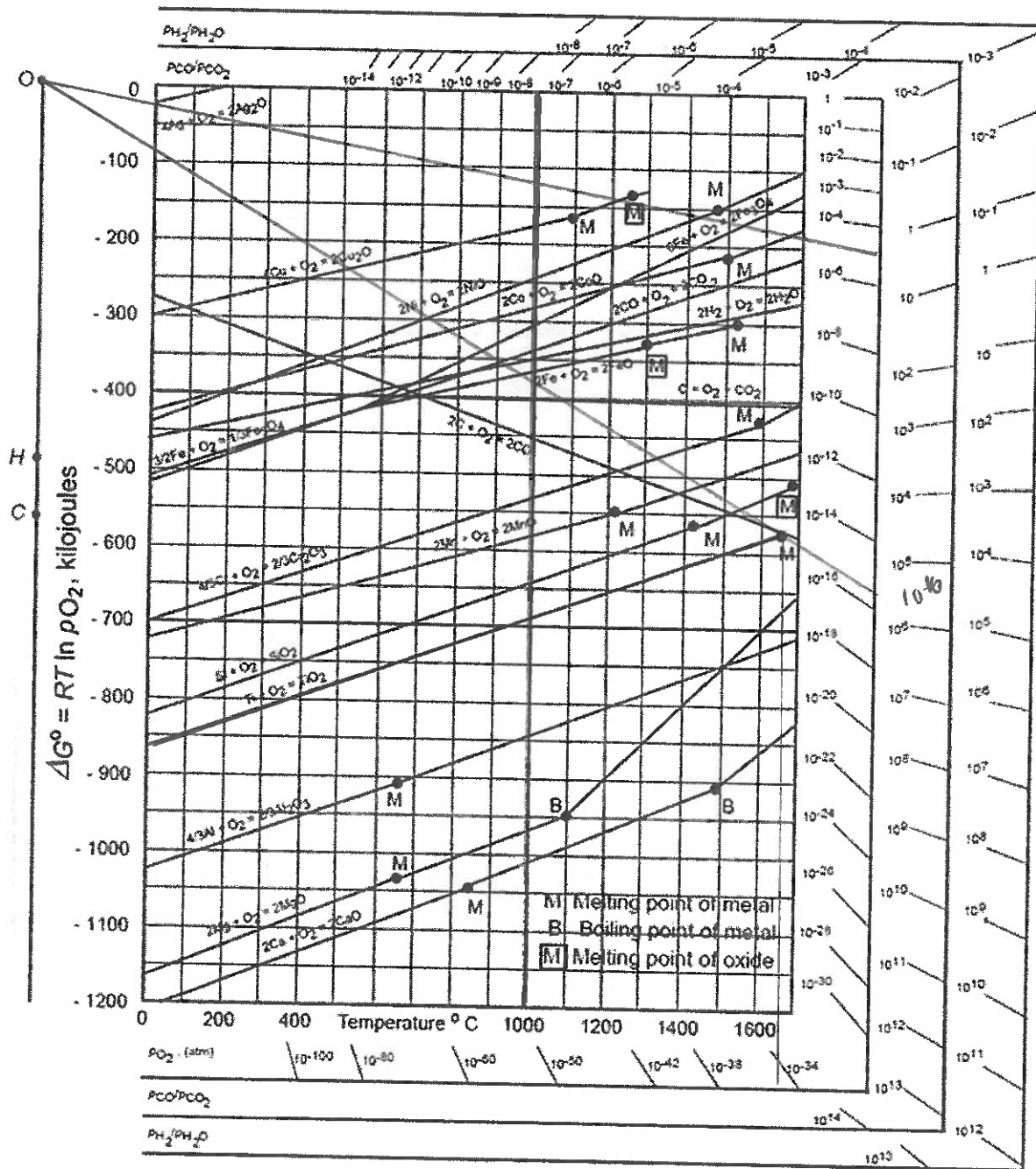
ANS: A

(x) In a dilute solution, Raoult's Law states that.

- A activity coefficient of the solute is approximately equal to 1
- B activity coefficient of the solvent is approximately 1
- C activity coefficient of the solute is constant
- D activity coefficient of the solute is approximately equal to 1
- E activity coefficient of the solute = activity coefficient of solvent

ANS:

B



1650°C